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January 29, 1974
SUPERSEDING
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May 15, 1964

FEDERAL SPECIFICATION

CLEANING AND POLISHING COMPOUND, FLOOR (PASTE EMULSION)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a polishing and cleaning compound which is intended primarily for use on wood and cork tile floor surfaces (see 6.1).

1.2 Classification. This specification covers one grade of material.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to extent specified herein:

Federal Specifications:

PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 through 12 gallon).
PPP-P-001655 - Pails, Plastic, Shipping and Storage.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society For Testing and Materials (ASTM):

D 217 - Method of Test for Cone Penetration of Lubricating Grease.
D 2047 - Static Coefficient of Friction of Polish-Coated Floor Surfaces as Measured by the James Machine.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material characteristics.

3.1.1 Penetration. The cone penetration shall be in the range of 260-350 tenths of a millimeter when tested according to 4.4.1.

3.1.2 Total solids. Total solids (nonvolatile matter) shall be 24 ± 1 percent by weight when tested according to 4.4.2.

3.1.3 Dispersability. The compound shall be completely dispersable in water to form an emulsion which shall show no separation or sedimentation after 2 hours when tested according to 4.4.3.

3.2 Performance characteristics.

3.2.1 Gloss. The 60-degree specular gloss shall be not less than 10 when tested according to 4.4.4.

3.2.2 Slip-resistance. The slip-resistance shall be not less than 0.50 when tested according to 4.4.5.

3.2.3 Appearance. The material shall provide a uniform wax film which will not discolor the floor surface when tested as specified in 4.4.6.

3.2.4 Ease of application. The material shall spread without difficulty, and shall buff evenly when applied in a thin coat upon the floor, when tested as specified in 4.4.6.

3.2.5 Cleaning ability. The material shall remove substantially all the dirt embedded in the old floor wax when tested as specified in 4.4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, unless otherwise specified. For purposes of sampling for inspection, a lot shall consist of all material submitted for inspection at one time.

4.3 Inspection.

4.3.1 Inspection of material and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3.2 Inspection and test of the end item.

4.3.2.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled primary containers of compound of the same grade for examinations under 4.3.2.1.1, 4.3.2.1.2 and 4.3.2.1.3, and in units of shipping containers for examination under 4.3.2.1.5.

4.3.2.1.1 Examination of the end item container for defects in closure, appearance, markings, construction and workmanship. The sample unit for this examination shall be one filled primary container.

Examine	Defect
Container	Not specified type, size or style container. Any evidence of leakage or seepage from any part of the container. Any hole, tear, puncture, scuff, dent or other defect in body of container. Closures not as specified. Interior and exterior coating (when applicable) not as specified.
Labeling	Printing or stenciling pertaining to the applicable information and instructions for end item use of the specified type material omitted, illegible, incorrect, incomplete, smeared.

4.3.2.1.2 Examination of the end item container for defects in appearance of compound. The sample unit for this examination shall be one filled primary container.

Examine	Defect
Appearance of compound	Presence of dirt or foreign matter. Not homogeneous. Form not as specified. Objectionable odor.

4.3.2.1.3 Examination of the end item container for defects in net contents. The sample unit for this examination shall be one filled container. The average net contents shall be not less than the specified or indicated quantity.

4.3.2.1.4 Testing of the end item. Sixteen ounces of material shall be drawn from each of five shipping containers and composited. The test sample shall consist of a thirty-two-ounce composite. Unless otherwise specified, duplicate determinations shall be performed on the composite with no evidence of failure to meet the specified requirements.

4.3.2.1.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table I. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the

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closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be II and the AQL shall be 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container. Bulging or distortion of container.

4.3.2.1.6 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection level for determining the sample size, and the acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.2.1.1	S-6	2.5
4.3.2.1.2	S-5	1.5
4.3.2.1.3	S-4	---
4.3.2.1.5	S-4	4.0

4.4 Test procedures.

4.4.1 Penetration. Penetration shall be tested according to ASTM D-217.

4.4.2 Total solids. For total solids, weigh to the nearest milligram, approximately two grams (g) of the sample, in duplicate, into tared, dried, flat bottom dishes of approximately six centimeters (cm) in diameter. Place in a forced draft drying oven and heat at 105-110°C for 16 hours. Cool, weigh, and calculate results. Duplicate samples shall check within 0.1 percent and the average shall be within the concentration specified.

4.4.3 Miscibility with water. To test for miscibility with water, 50 grams (g) + 2.5 g are placed in 500 cc of distilled water and agitated with a stirrer for 10 to 15 minutes. The paste emulsion shall be readily dispersed into a uniform emulsion of liquid consistency. Non-miscibility is readily apparent with the solvent paste waxes in contrast to the emulsion finishes. Any separation of the emulsion within a period of 2 hours shall be cause for rejection.

4.4.4 Gloss. Two coats of wax are applied to sealed duplicate planed oak wood panels. The surface is left smooth with the applicator to obtain a better polished surface with a minimum of effort. The first coat is dried for at least two hours and is polished with a No. 00 steel wool pad. A second uniform coat of wax is applied. The second coat is dried for 18 hours. It is also polished with the steel wool pad. Gloss measurements are made with a 60-degree specular gloss meter upon the second polished coat.

4.4.5 Slipperiness. Duplicate oak wood panels which have been buffed to a high polish are used in the determination of the slipperiness factor in accordance with the method described by ASTM Method D-2047. The panels shall be aged for not less than 24 hours prior to testing.

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4.4.6 Appearance, Application and Cleaning. Select a soiled, waxed wooden floor. Apply the wax to about half of the floor 1/, and polish using a standard machine with a 17" or a 19" disc. Note the spreading, cleaning, and buffing properties.

4.4.6.1 Soiled floors. Ideally, the test floor should be at or near the end of the normal maintenance cycle. If such a floor is not available, a clean floor may be soiled for testing by spreading 6 to 8 fluid ounces of CSMA Standard Soil 2/ over 20 square feet and buffing with a polishing pad for several minutes. Any excess soil should be removed by sweeping.

1/ In the event that future tests in the same area are anticipated, it is suggested that the entire floor be cleaned when testing has been completed.

2/ Available from Gardner Laboratory Inc., 5521 Landy Lane, Bethesda, MD 20016.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The material shall be in a pail conforming to PPP-P-704, type II, class 3.

5.1.2 Level B. The material furnished in 5 gallon quantities shall be packed as described in 5.1.1, or in a pail conforming to PPP-C-001655.

5.1.3 Level C. The specified quantities (see 6.2) of the material shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall be in compliance with Uniform Freight Classification or National Motor Freight Classification as applicable.

6. NOTES

6.1 Intended use. The product is to be used without dilution and applied by the use of a steel wool pad under a concentrated weight polishing machine or other suitable means. It is intended only for use on wood or cork tile. As this product is presumed to contain petroleum solvents, it should only be used with extreme care on other types of tile.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Quantity of compound desired.
(c) Levels of packing required (see 5.1).

MILITARY INTEREST:

Army - MQ
Navy - SH
Air Force - 84

Preparing Activity:

GSA - FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

VA - HEW

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain copies and other documents referenced herein. Price 10 cents each.

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